Measuring School Poverty: An Exercise in Convergent Validity

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Abstract

Many educational policies hinge on the valid measurement of student poverty at the school level. However, recent research raises questions about widely used measures based on free and reduced-price lunch (FRPL) enrollment. Using unique data linking individual students to IRS tax records and other data housed at the U.S. Census Bureau, we provide the first comprehensive examination of how well different measures capture school poverty. We find that direct certification data provide the best widely available measure of poverty, both over time and across the distribution of school poverty. FRPL data quality varies, suggesting that in some contexts FRPL-based measures – including measures from the Urban Institute – perform relatively well. By contrast, neighborhood-based measures perform relatively poorly.

Keyword*:* School Lunch; School Poverty; Measurement of Poverty; Validity; Low-income Students

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We are living through an era of widening economic inequality as well as widening gaps in academic achievement and educational attainment between poor and non-poor students (Reardon 2011; Bailey and Dynarski 2011; Duncan, Kalil, and Ziol-Guest 2017; Ziol-Guest and Lee 2016). This historical context has brought renewed interest to long-standing efforts to improve the educational resources and opportunities available to students growing up in poverty. Many such compensatory strategies, including Title I of the federal *Elementary and Secondary Education Act* as well as a variety of state and district-level programs, earmark supplementary funds and educational programs to schools and districts that educate relatively large proportions of poor and near-poor students (Greenberg, Blagg, & Rainer, 2019; U.S. Department of Education, 2019).

To operate efficiently and equitably, compensatory school funding programs require valid measures of the concentration of student poverty at the district and school levels (Marar, 2020). However, substantial questions exist about the degree to which the standard existing measure – the proportion of students enrolled in means-tested free or reduced-price lunch (FRPL) via the National School Lunch Program (NSLP) – can accurately capture poverty in schools (Domina et al. 2018; Michelmore & Dynarski 2017; Harwell & LeBeau 2010).

In this paper, we assess the validity of FRPL as a measure of poverty in schools and compare it with other potential measures. Specifically, we assess the following policy- and research-relevant measures of school poverty: two different sources of FRPL enrollment rates, so-called direct certification rates (which measure of the proportion of students who are enrolled in SNAP and other means-tested programs targeted at low-income families), aggregated data describing the rate of poverty in the neighborhoods in which students live, and experimental measures from the Urban Institution and the National Center for Education Statistics. To do so,

we link data describing all students enrolled in Oregon public schools to confidential federal tax return and program participation microdata. We use tax information and program participation data to construct a benchmark school poverty measure that captures the proportion of students enrolled in each Oregon public school whose family either reported a household income of less than 185 percent of the federal poverty level or enrolled in SNAP in each study year. We posit that any measure used to distribute school finances aimed at compensating for school poverty ought to mirror variation in poverty rates over time and across schools in an unbiased fashion. As such, our analyses compare temporal trends for our benchmark measure and available candidate measures; examine the cross-school correlations between these measures; and focus on the extent to which the benchmark and candidate measures correspond across the distribution of school poverty.

Our analyses reveal that many widely available measures of school poverty correlate reasonably highly with our school poverty benchmark measure. However, each of these measures is subject to important shortcomings. We find that the validity of widely used FRPL enrollment rate depends on how these data account for growing number of schools that participate in universal free meals programs, the most notable of which is the Community Eligibility Provision. By contrast, direct certification rates correlate highly with benchmark poverty rates. Not all states make this measure available for public use and fewer use it for policy purposes. But our analyses suggest that it provides the most valid currently available measure of poverty at the school level.

Other existing alternatives, including measures that use population-based estimates of household poverty rates or income levels of households living near students or their schools, are

less promising. Such measures also preclude understanding which specific students are in poverty, which, though not a focus of our analyses, is an important feature for research studies.

Defining the construct: "Students from low-income families"

Our focus in this paper is on the valid measurement of student poverty rates at the school level. While other measures of student disadvantage and academic need are clearly important, we focus on poverty because of the central role that it has long played in U.S. educational policy. Over the last several decades, policymakers at the federal, state, and local levels have authorized a wide array of policies that direct supplementary funds to public schools and districts based on the concentration of poverty among enrolled students.[2] These policies conceptualize poverty as a unidimensional and dichotomous construct – a student is either poor or not, based on their household income. In line with this policy tradition, we focus largely on school poverty measures constructed from dichotomous indicators of student poverty based on income.

The largest and most influential policy aimed at providing additional support for poor students is Title I, Part A of the 1965 *Elementary and Secondary Education Act* (ESEA). Title I allocates supplemental federal educational funds to "local educational agencies serving areas with concentrations of children from low-income families to expand and improve their educational programs." The policy's language notes the "special educational needs of children of low-income families and the impact that concentrations of low-income families have on the ability of local educational agencies to support adequate educational programs" (Public Law 89-10-Apr. 11 1965, p. 27).

[2] Throughout this paper, we use the term "poverty" as a shorthand for the condition described by the phrase "students from low-income families." At several points in the manuscript, we make specific reference to the federal poverty line, a construct that is defined around a family's ability to secure a minimal standard of living. Since this construct is dated and does not account for regional variation in costs of living, scholars debate its validity (Hauser, 1994). Our discussion does not take a stand on this issue, but our use of the term poverty is meant to reflect its colloquial use rather than the official federal standard, unless we explicitly refer to the federal poverty line.

Currently 35 states also provide additional funding for pre-primary, elementary, secondary and higher education targeting low-income students (Edbuild, 2022). Furthermore, many of the nation's largest districts have implemented weighted student funding formulae that inform the distribution of district resources across schools. While existing district-level systems consider a range of student attributes, including the concentration of English Language Learners and special education students, most include student poverty as a key factor for targeting compensatory resources (Roza, Hagan & Anderson 2021). These federal-, state-, and district-level policies vary in many regards. Different policies operationalize the notion of economic disadvantage or poverty in different ways, with some emphasizing the federal poverty line and others emphasizing different income-based thresholds. Policies also differ in the degree to which they supplement allocations based on student poverty or other forms of economic disadvantage and the range of other characteristics that inform resource allocations (EdBuild, 2022; Roza, Hagan & Anderson 2021). Nevertheless, common to each of these policies is the notion that school finance mechanisms can offset disparities associated with socioeconomic inequality by directing funds toward the schools and districts that educate students from low-income families.

Measures of school-level economic disadvantage

In this section, we provide an overview of strategies currently being used for the measurement of student poverty at the school level in U.S policy contexts as well as several promising alternative strategies.[3]

[3] Before summarizing these school-level measures, we note that under Title I of the ESEA, the federal government allocates supplementary funds to districts based on data generated by the U.S. Census Bureau's Small Area Income and Poverty Estimates Program (SAIPE). SAIPE uses state and county-level American Community Survey (ACS) data as well as summary data from federal income tax returns, SNAP benefits data, data from the decennial Census, and other sources to estimate the percent of children ages 5-17 living in the district's geographic boundaries whose household income falls below the poverty line using (U.S. Census Bureau 2021). Since these measures are largely derived from data reported by families sampled to participate in the ACS, they are not available at the school level. As such, the SAIPE data are beyond this paper's analytic scope.

FRPL enrollment

To measure poverty at the school level, policymakers typically rely on data produced in the administration of means-tested programs designed to benefit young people living in poor or near-poor households. Historically, the most widely used measure is based on the proportion of students enrolled in FRPL, often referred to by its acronym as the FRPL rate. Currently approximately 30 US states use FRPL data to measure school poverty (EdBuild, 2022).

Scholar and practitioners have long known that FRPL rates provide error-prone measures of school economic disadvantage (Domina et al. 2018; Harwell and LeBeau, 2010). Historically, students enroll in FRPL when their households self-report income at or below 185 percent of the federal poverty line. This self-reporting process produces two-sided error. Some students from relatively high-income families enroll in FRPL; other students from low-income families do not enroll (Domina et al. 2018). Since administrative burdens associated with completing applications, concerns about being stigmatized as poor at school, and governmental distrust likely vary across schools and communities (USDA, 2022, Richman, 2019, Domina et al. 2018; ERS, 2006; Harwell and LeBeau, 2010), the measurement error associated with school-level poverty measures based on aggregated FRPL data likely varies by school location and enrollment demographics.

Recent efforts to improve student access to meals may have further exacerbated concerns regarding the ability of FRPL data to measure economic disadvantage (Cookson, 2020; Koedel & Parsons 2021). Specifically, Provisions 1, 2, and 3 of the National School Lunch Program (implemented in the 1980s as alternative provisions to the normal requirements for annual determinations of eligibility for FRPL) and the Community Eligibility Program (CEP, implemented nationally in 2015) permit schools to provide free lunch widely, and directly or

indirectly reduce data collection obligations. During the 2019-20 school year, approximately 33 percent of all K-12 public schools participated in CEP (FRAC, 2020). Some schools continue to report traditional FRPL enrollment data after enrolling in CEP, while others stop reporting or report that all students are enrolled (Greenberg et al. 2019). The pandemic potentially brought additional changes to FRPL enrollment data as federal pandemic relief efforts allowed schools across the country to offer free meals to all students beginning in the spring 2020. While this program ended in the fall of 2022, it is not clear when and to what extent schools and districts will resume FRPL data collection.

Direct Certification

In an attempt to reduce the administrative burden associated with enrolling in FRPL, the USDA authorized districts to directly certify children from households enrolled in the Supplemental Nutrition Assistance Program (SNAP), Temporary Aid for Needy Families (TANF) or other means-tested programs for free lunch beginning the 1990s and early 2000s.

Many states and school districts now report data documenting the proportion of students who enroll in FRPL via direct certification and use direct certification data either as a supplement to or a replacement for traditional FRPL rates for resource allocation. To date, limited evidence is available regarding the properties of direct certification as a measure of school poverty. These data are not strictly comparable to FRPL rates, since the criteria for receiving benefits via SNAP and TANF are more restrictive than the criterial for enrolling in FRPL via the traditional self-report mechanism.[4] Furthermore, direct certification rates have both potential strengths and weaknesses relative to NSLP data as a measure of school poverty. Since families must provide

[4]Families must demonstrate household income less than 130 percent of poverty and have less than $2250 in cash savings to enroll in SNAP. TANF eligibility rules vary across states, but are typically more restrictive than SNAP eligibility rules.

income documentation to enroll in SNAP and TANF, direct certification rate data may be less subject to some forms of bias than FRPL enrollment data. At the same time, the policies and administrative practices that govern the process of enrolling in SNAP, TANF, and Medicaid vary across states, within states, and over time (Ganong and Liebman, 2018; Dickert-Conlin et al., 2021). This variation likely leads to both geographic and temporal variation in the characteristics of families who are captured in direct certification. In particular, direct certification rates may substantially understate poverty rates in contexts where program enrollment processes place heavy burdens on potential beneficiaries relative to contexts where administrative burdens associated with program enrollment are less pronounced (Herd & Moynihan 2018; Heinrich et al. 2022). Furthermore, direct certification data may understate poverty in schools serving immigrant, rural, and other communities in which income-qualified families are relatively less likely to enroll in means-tested programs.

Newly introduced alternative measures

In order to address the shortcomings associated with measures of economic disadvantage based on program participation data, researchers and policymakers have begun to experiment with alternative approaches. Some states and districts have begun to construct school-level measures of student neighborhood poverty using ACS data. While these data do not directly measure the household income of students enrolled in schools, they use student home address data to highlight the Census block groups or tracts in which students live to measure the economic conditions in students' neighborhoods. Assuming that students' household socioeconomic conditions correlate highly with their neighbors' household socioeconomic conditions, aggregating these measures to the school level may be a useful proxy for the socioeconomic conditions of students enrolled in schools.

The Urban Institute recently developed a measure of school poverty that builds on two sources of publicly available data: the school-level direct certification or FRPL data available via the National Center of Education Statistics' Common Core of Data (CCD) and the Census-constructed district-level Small Area Income and Poverty Estimates Program (SAIPE). This measure, known as Model Estimates of Poverty in Schools (MEPS), uses a linear probability model to adjust school-level estimates of economic disadvantage based on average household poverty rates for families with children in the districts in which schools are located. MEPS data potentially address error associated with FRPL or direct certification data, based on two assumptions: (1) that school-level biases in FRPL or direct certification data cluster at the district level (as opposed to varying across schools within the same district), and (2) that SAIPE data capture public school student poverty rates at the district level, despite being based on a sample that is not designed to specifically generalize to the subset of families enrolling children in public schools. SAIPE data captures proportions of school-aged children below 100 percent of the federal poverty line within a school district, which is a lower threshold of poverty than other typical definitions (Gutierrez, Blagg, and Chingos, 2022). Because of this, MEPS identifies fewer students as poor compared with other measures.

The National Center for Education Statistics (NCES) reports School Neighborhood Poverty (SNP) statistics. This measure uses ACS data to estimate the average household income relative to the poverty line for the households with school-aged children who live in close physical proximity to each school's campus (Geverdt and Nixon, 2018). Recent work by Fazlul et al. (2021) suggests a transformation of the SNP measure designed to better match the dichotomous conception of poverty embedded in policy, capturing the share of students at or below 130 percent of the federal poverty line for each school. Based on the assumption that the

economic circumstances of households located near a school's campus are representative of the economic circumstances of all the households in which enrolled students live, these SNP-based measures have been proposed as a proxy for school poverty (Fazlul et al. 2021). This may be a problematic assumption, particularly for schools that serve geographically dispersed student bodies.[5]

Current study

Our analyses investigate the validity of various measures of school poverty, with the goal of identifying measures that are both reliable and widely available for use in educational policy and practice. Our argument-based approach to validation (Kane, 2006, 2012) proceeds by establishing a benchmark measure and then assessing alternative measures against our benchmark.

Specifically, our analyses address the following research questions:

1. To what degree do available measures track with temporal trends in poverty for students enrolled in Oregon public schools?

2. To what degree do available measures capture variation in poverty across Oregon public schools, and how has their correlation with poverty changed over time?

3. To what degree do available measures capture variation across the distribution of school poverty? In particular, do candidate measures over- or under-estimate school poverty rates in schools enrolling high or low concentrations of students from low-income families?

[5] Through an additional program called BlindSIDE, NCES has partnered with states to provide policymakers with a spatially interpolated school poverty measure based on student addresses, rather than school addresses (Institute of Education Sciences, 2021). While we are unable to assess the properties of this proposed measure here, we note that it has a greater degree of face validity than the existing SNP measure. In FY2019, multiple State Departments of Education received additional funds from the State Longitudinal Data Systems Grant Program to help the National Center for Education Statistics to test this new Spatially Interpolated Demographic Estimates (SIDE) poverty measure. States participating in the testing of the SIDE measure collected and cleaned student addresses and obtained poverty estimates using the SIDE program and compared this measure with existing measures of poverty and other academic indicators, including FRPL data and test scores.

Data

We link student-level administrative records from the Oregon Department of Education (ODE) for the school years 2009-10 through 2016-17 with administrative records stored at the U.S. Census Bureau. These data are linked using protected identification keys (PIKs) assigned via the Census's Person Identification Validation System.[6] We link students in the ODE data to 1) the Oregon SNAP enrollment records for each school year and 2) the IRS form 1040 records (tax returns) in which they are claimed as dependents for the most recent and two prior calendar years (spanning from 2008 to 2017). Details about measures used in the analyses are included in Table 1 and described below.

Benchmark Measures of School Poverty

Benchmark (SNAP/IRS 3 year). Our benchmark measure of school poverty defines students as low-income in a given year if their family either enrolls in SNAP at any time during the given school year or has an adjusted gross income that is less than 185 percent federal poverty in the most recent calendar year (i.e., income for 2016 in the 2016-17 school year). For students who are not enrolled in SNAP or claimed on a 1040 tax form in the most recent calendar year, we check 1040 records from the prior two calendar years, and use this income information.[7] Our benchmark is interpreted as the proportion of students who are low-income, with 185 percent federal poverty as our target threshold.[8]

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[6] For a detailed description of the Person Identification Validation System, see Wagner and Layne (2014). We are able to link 96% of K-12 students in the ODE data to PIKs. We exclude unlinked students from our calculations of school-level economic disadvantage rates. Broadly speaking, students might not be assigned a PIK due to missing or erroneous information in administrative records for the student, or because the student does not have a social security number at all, which means a PIK has never been created for them.

[7] We find that 6% of Oregon students do not have a low income value, either because they do not have a PIK or because they do not appear on a 1040 record within the three years prior to the reference year and they do not receive SNAP at any point during the reference year. Our investigation of students missing this information did not support the assumption that all of these students are low income, thus we proceed with the implicit assumption that they are missing at random.

[8]SNAP eligibility falls at 130% federal poverty, but students occasionally only briefly enrolled in SNAP during the

Benchmark (IRS 1 year). We construct an alternative version of our benchmark measure that uses only IRS income data from the most recent calendar year to measure the proportion of students with household income below 185 percent of federal poverty. This measure systematically excludes students whose families did not recently file tax returns, but it offers a measure that is independent of program participation, and one that states may be able to more easily replicate with state level tax data. It also provides a more useful benchmark for comparing the performance of a SNAP-based measure to other candidate measures.

Traditional FRPL Measures

Two distinct measures of school-level poverty rates based on FRPL enrollment are currently available for Oregon public schools. Our analyses use both of these measures, since they highlight potentially important differences in how FRPL rates address measurement questions associated with CEP and other programs that offer free school meals to all students at participating schools.

CCD FRPL. The NCES Common Core of Data (CCD) collects FRPL rates for Oregon public schools, which we refer to as "CCD FRPL".

ODE Economic Disadvantage. The ODE data flags students as "economically disadvantaged" based on their FRPL enrollment and other factors, including homeless children, as determined by the McKinney-Vento Homelessness Assistance Act (enacted in 2001). In 2015, the ODE data began to flag students as "economically disadvantaged" if they attend schools that provide free lunch to all regardless of economic status. Thus, our "proportion economically

school year may have income in excess of 185% federal poverty. However, our analysis suggests that these students overwhelmingly fall below the 185% threshold, so this measure still targets that threshold though it may not perfectly comply with it.

disadvantaged" variable is the proportion of students in each school designated economically disadvantaged in the ODE microdata.

Direct Certification Measure

SNAP. While many states have started publishing school-level direct certification rates, Oregon did not during the time period of study. Thus, we construct an estimate of the direct certification rates for Oregon schools using Oregon SNAP data housed at the U.S. Census Bureau. Most of the SNAP administrative data that the U.S. Census Bureau possesses have been acquired since 2012 as part of the *Census-FNS-ERS Joint Project*, a cross-agency collaboration between the State SNAP agencies, the U.S. Census Bureau, and USDA's Food and Nutrition Service (FNS) and Economic Research Service (ERS).[9] This measure is simply the proportion of students at each school that are enrolled in SNAP at some point during the school year.[10]

An important distinction between our direct certification proxy and actual measures of direct certification is that different states have different methods for matching students to the enrollment records for whichever public program they use to administer direct certification. The quality of the matching approach may affect the quality of the measure in a manner we are not able to assess.

Alternative Candidate Measures

ACS Block Group Measure. We construct an ACS-based measure by linking students in each school to their residential block group, then assigning them the low-income rate (proportion

[9] For more information about the *Census-FNS-ERS Joint Project,* see the webpage: https://www.ers.usda.gov/topics/food-nutrition-assistance/food-assistance-data-collaborative-research-programs/snap-and-wic-administrative-data/

[10] Using SNAP enrollment directly instead of state-reported direct certification data avoids potential issues that may arise from not all eligible students being directly certified. In school years 2017-18 and 2018-2019, the USDA estimates that 98 percent of SNAP-participant children were directly certified for free school meals. This was lower in Oregon, where 88 percent of children enrolled in SNAP were directly certified for free school lunch (USDA, 2021).

of households with income less than 185 percent of the federal poverty line according to the ACS 5-year estimate) of their block group.[11] We average these low-income rates across students to arrive at a school-level poverty measure.

Urban Institute MEPS. We utilize the publicly available measures available on Urban Institute's website, including the Model Estimates of Poverty in Schools (MEPS) and the modified MEPS. The Urban Institute suggests that analysts use the original MEPS when looking at schools of varying enrollment sizes. We follow the Urban Institute's suggestion and report findings from the original MEPS in our main analyses and provide information about the modified MEPS in footnotes.

NCES-SNP. As described above, SNP measures the household income-to-poverty ratio for ACS respondent households living near schools. This measure is not available prior to the 2016-17 academic year.

Transformed NCES-SNP. Fazlul et al (2021) propose a transformation of the SNP designed to better match the dichotomous conception of poverty embedded in policy and other widely used measures. We reproduce this transformation and report results of this transformed variable as well as the untransformed version.[12]

[TABLE 1 AROUND HERE]

Sample of Schools

Table 2 provides descriptive statistics for the analytic sample. To construct a panel of schools comparable across all years of our study, we make an analytic sample of schools that (a)

[11] There are different potential options for translating ACS income and poverty measures into school poverty rates. We found this approach superior to designating students in a binary fashion as either low or high income based on their neighborhood low-income rate or neighborhood average income.

[12] To do so, we first multiplied the provided standard errors of the SNP measure by five and calculated the proportion of the area under the normal curve that lies below 185 percent federal poverty, given the SNP mean and standard error.

appear in ODE administrative data in all 8 years of our study, and (b) are available for all our candidate measures. All the analyses in this paper are restricted to the approximately 1,060 schools in our analytic sample. We lose approximately 300 schools with small enrollments driven in large part by instances for which there is no school-level data available in NCES data sources like the CCD and SNP. Our analyses consider only measures that are widely available throughout the state, but we note that some measures are not available for all schools. There are a small number of low-enrollment Oregon public schools in which no enrolled students are linked to data with which we construct the benchmark measure (they may be missing a PIK or, if not, do not appear in form 1040 or SNAP records). Still, the analytic sample offers good coverage of the universe of schools in the ODE data (see Appendix Table 1). Limitations in data availability may matter more substantially in practice, where missing data can restrict efforts to allocate compensatory funds.

[TABLE 2 AROUND HERE]

Methods

We take a descriptive approach to answer our three research questions which we describe in this section. To answer our first research question, below we plot state-wide trends in school poverty for Oregon public schools on all available measures between 2010 and 2017.

To answer our second research question, we correlate all the candidate school poverty measures with our benchmark separately for each school year. Correlation coefficients indicate how well the various measures preserve the relative rankings of schools according to our benchmark. In light of concerns about the way changes in policy and practice induce change in proxy measure performance, we further assess change in correlation coefficients over time for

each measure that is available in multiple years. Usefully, correlations are unaffected by differences in the level of school poverty targeted by different measures.

However, a correlation is a simplistic summary of the relationship between two variables, and in particular, correlation analyses may overlook systematic divergences between measures in very low- or high-income schools. Our third research question draws attention to the performance of candidate measures across distribution of school poverty. To do so, we focus on our most recent year of data (2016-2017). We standardize each of our measures, group schools into 20 quantiles based on their rank according to the benchmark measure,[13] and calculate the mean and standard deviation of the candidate measures for schools in each quantile. We then plot these binned statistics. Each point on this plot represents the mean for the candidate measure in each quantile bin, providing a visual summary of how consistently the benchmark and candidate measures agree about the relative poverty of schools. Each of the whiskers represents the standard deviation of the candidate measures in each quantile bin, providing a visual summary of the extent of variation in the candidate poverty rates in each bin.

Results

Temporal trends in measures of student poverty

Figure 1 provides an overview of each of the available measures of poverty in Oregon public schools between the 2009-10 and 2016-17 school years. Since these measures are enrollment-weighted, they can be interpreted as the proportion of Oregon public school students designated as poor. Since the six measures represented on this graph use different income thresholds to define student poverty, it is not surprising that these lines represent varying levels.

[13] These quantiles are enrollment-weighted so each bin is comprised of schools enrolling approximately 5% of students.

However, we generally expect the measures to follow similar trends; as the share of students receiving SNAP falls, so too should the share enrolled in FRPL and the share designated in poverty by MEPS.

[FIGURE 1 AROUND HERE]

The solid line near the top of the graph represents the benchmark (IRS/SNAP 3 year) measure, which is the statewide trend in the proportion of Oregon public school students whose family reported household income less than 185 percent of poverty in IRS filings and/or enrolled in SNAP. This rate increases modestly at the beginning of the study period as the state slowly emerged from the 2008 recession, peaking at approximately 52 percent before declining to approximately 47 percent in 2017.

The dashed line with the circle marker in this graph represents the trend in FRPL enrollment rates, as reported by the state of Oregon to the Common Core of Data; the dashed line with a triangle marker represents trends in the Oregon Department of Education's economically disadvantaged measure. The contrast between these two measures – both of which are ostensibly built around FRPL enrollments – is instructive. While the CCD-reported trend largely parallels the trend in our benchmark measure, the economically disadvantage rate – which counts schools that participate in CEP as 100 percent economically disadvantaged – diverges from this trend, rising markedly as schools across the state implement CEP.

As expected, the remaining measures – SNAP enrollment, ACS neighborhood poverty, and MEPS – record very different levels of poverty than the benchmark. SNAP enrollment rates, which reflect students whose families demonstrate household income below 130 percent of poverty, vary between 32 and 37 percent across the time period; ACS neighborhood poverty rates range similarly from 30 to 35 percent; and the MEPS rate, which reflects students whose families

have household income below 100 percent of poverty, varies between 20 percent and 15 percent in the years for which it is available. Trends for these three measures largely parallel the benchmark's trend over time.

Correlations between the benchmark measure and candidate measures

Table 3 summarizes the school-level correlations among the benchmark (IRS/SNAP 3 year) and each of the traditional FRPL measures, direct certification measure, and alternative candidate measures described above in the 2016-17 school year, the most recent year for which we have data. Despite widespread concerns about the validity of traditional FRPL enrollment measures as measures of student poverty, our analyses indicate that the CCD FRPL measure correlates well with the benchmark in 2017, at 0.934. However, the ODE Economic Disadvantage measure, which is largely based on FRPL enrollment, has a lower correlation with the benchmark of 0.863.[14] The difference in the correlation between the two traditional FRPL measures and the benchmark indicates that the quality of measures capturing FRPL enrollment is contingent on the decisions that data providers make as they construct and report data.

The analyses in Table 3 show that the direct certification measure, constructed using SNAP enrollment, correlates extremely highly with the benchmark at 0.957.[15] Concerns about the measurement quality of school poverty measures based on free and reduced-price lunch enrollment have also led some to consider the use of alternative measures. Our ACS Block Group measure, which describes the degree of economic disadvantage in the neighborhoods in

[14] We additionally constructed a persistent economic disadvantage measure based on Michelmore and Dynarski (2017) definition, flagging students as economically disadvantaged if they were economically disadvantaged for the prior 3 years. This persistent economic disadvantage measure correlates at .835 with the benchmark.

[15]Since SNAP enrollment data contribute to our benchmark measure, some of this correlation is mechanical. However, an alternative version of our benchmark measure based exclusively on the proportion of students whose families report income below 185 percent of the poverty line in their tax filings also correlates very highly with the proportion enrolled in SNAP measure, a correlation of 0.942.

which schools' students reside, correlates with the benchmark at 0.796. The Urban Institute's MEPS, which is based on CCD, correlates highly with the benchmark similarly to the CCD FRPL measure at 0.935.[16] Meanwhile, the NCES-SNP measure, which captures the rate of poverty for families living in schools' immediate vicinity, is -0.694. The negative correlation is expected as the measure is an income-to-poverty ratio which is naturally inversely related to proportion low income. A transformed NCES-SNP measure (cf, Fazlul et al. 2021) has a correlation of 0.661. Taken together, the results in Table 3 indicate that alternative candidate measures that use data from the ACS correlate less well with the benchmark than do more readily available direct certification or FRPL enrollment-based measures.

[TABLE 3 AROUND HERE]

Figure 2 graphs correlations between the benchmark, traditional FRPL-based measures, direct certification measure, and alternative candidate measures over time. We see that the quality of traditional FRPL-based measures varies over time. The correlation between the benchmark and the CCD FRPL measure improves across the study period, while the correlation between the benchmark and the ODE Economic Disadvantage measure rises and then falls over time. While we do not have an explanation for the improvement that we observe in the ODE Economic Disadvantage measure's correlation with the benchmark between 2010 and 2014, we note that the post-2014 decline corresponds with the CEP's implementation in schools across the state.

Encouragingly, this figure indicates that our proxy for direct certification – the measure of the proportion of students enrolled in SNAP (the dashed line with the square marker) – consistently correlates extremely highly with the benchmark in each of the available years. If

[16] The modified MEPS measure has a lower correlation with the benchmark than the original MEPS, with a correlation of .865.

anything, this measure appears to improve in quality over time, with a correlation of .944 in 2010 and .957 in 2017.

We note that although the ACS Block Group measure does not correlate particularly highly with the benchmark at any point in the study period, its correlation is relatively stable over time. While we only have access to data on the MEPS measure for the last five years in the study period, we note that this measure appears to be somewhat more stable than both of the traditional FRPL enrollment-based measures.

[FIGURE 2 AROUND HERE]

Relationship between benchmark measure and candidate measures across the distribution

Figure 3 presents a series of binned scatterplots depicting the relationship between the benchmark and candidate measures across the poverty distribution. These graphs represent data from the 2016-17 academic year. We use z-scores to standardize all measures, allowing us to focus on the ranking of measures as opposed to the levels of poverty they suggest and thus effectively setting all measures on the same scale. For purposes of legibility, we separate these scatterplots into panels: Panel A depicts the traditional FRPL-based measures, Panel B depicts the direct certification measure (SNAP), and Panel C depicts the alternative candidate measures. Each point on these graphs represents a bin of 5 percent of schools (approximately 55 schools).[17] In the graphs, we plot bin averages of the standardized candidate measure of school poverty (on the y axis) for each bin of the standardized benchmark measure (on the x axis). We also mark one standard deviation above and below the mean of the binned benchmark measure, depicting how much variation each candidate measure has at different levels of the benchmark measure.

[FIGURE 3 AROUND HERE]

[17] We plot binned scatterplots due to privacy constraints with data housed at the U.S. Census Bureau.

The graphs in Panel A investigate the correspondence between the traditional FRPL-based measures and the benchmark, with the graph on the left focusing on the CCD FRPL measure and the graph on the right focusing on the ODE Economic Disadvantage measure. These traditional FRPL-based measures correspond closely with the benchmark across the distribution of student poverty, although the CCD measures poverty with greater precision across the distribution than the ODE Economic Disadvantage measure. At the high end of the poverty distribution, these measures appear to understate poverty relative to the benchmark measure, and this is particularly true for the ODE measure of economic disadvantage. This is due in large part to the fact that this measure reports all schools that participate in the CEP program – in 2017, 23 percent of schools in Oregon report a proportion of economically disadvantaged students greater than .99, while in the CCD FRPL-based measure only one school reports a proportion poor greater than .99. Because the ODE Economic Disadvantage measure reports all CEP schools at the scale's ceiling, this measure does not distinguish between those schools, and this dramatically alters the relative poverty rank of many relatively high-poverty schools.

While the graph in Panel B indicates that the SNAP-based direct certification proxy corresponds closely with the benchmark across the bulk of the distribution of school poverty, we note that the SNAP measure captures school poverty rates less precisely at the very top of the distribution compared with other points of the distribution. This suggests heterogeneity in SNAP enrollment in high poverty schools. Our data do not allow us to definitively account for why this might be, but SNAP under-enrollment in high-poverty schools is likely due to a relatively large concentration of children whose families qualify for SNAP benefits but choose not to enroll, a phenomenon that is particularly well-documented in immigrant communities (FRAC, 2020).[18]

[18] It is also possible that SNAP overcounts poverty in some schools due to provisions in SNAP enrollment that allow families to remain enrolled without recertifying every year or that over-enrollment might be an artifact of the timing

Also, the Food Distribution Program on Indian Reservations (FDPIR) may take the place of SNAP enrollment in schools near and in reservations. At the bottom of the distribution, we observe higher levels of the SNAP-based measure than expected based on the benchmark. This finding suggests that the relatively few students from low-income families in low-poverty schools have high levels of access to SNAP benefits.[19]

Looking at the alternative candidate measures in Panel C, we see that the ACS Block Group measure overstates poverty at the low end of the distribution and understates poverty at the high end of the distribution. Further, the ACS Block Group measure is relatively imprecise across the distribution of school poverty compared to other available measures. Finally, the Urban Institute's MEPS corresponds to the benchmark measure similarly to the CCD FRPL measure, which makes sense given that MEPS relies on CCD FRPL enrollment data. Like the CCD measure, the MEPS understates poverty in high-poverty schools. Unlike the CCD measure, the MEPS also slightly understates poverty in low-poverty schools.

Conclusion

As the distribution of income becomes increasingly unequal and achievement gaps between students from low-income families and their more affluent peers widen, a broad array of federal, state, and local policies aim to target resources and opportunities toward students from low-income families and the schools that educate them. Valid measures of school poverty are an essential component of this policy agenda.

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of income measurement.

[19] One might be worried that the high correspondence between the SNAP measure and the benchmark is attributable to a mechanical correlation driven by the use of SNAP to construct the benchmark measure. To assess this possibility, we looked at the correspondence of the SNAP based measure across the poverty distribution using an alternative benchmark that relies only on IRS data and find a similarly high degree of correspondence. Thus, we conclude that the high correspondence is not due to a mechanical correlation.

Our analyses compare several widely used school poverty measures as well as recently proposed alternatives against a benchmark measure constructed by linking student-level administrative data with IRS and SNAP data housed at the U.S. Census Bureau. Our findings build on a growing body of research highlighting the shortcomings of measures of school poverty that rely on FRPL enrollment information. Using data from Oregon public schools, we demonstrate that the proportion of students who qualify as economically disadvantaged according to the state's official measure has begun to diverge from the benchmark poverty rate as a growing number of schools begin to provide free meals to all students under the NSLP's Community Eligible Provision. Encouragingly, however, trends in FRPL enrollment data reported in the CCD continue to correspond to the benchmark. These CCD-reported FRPL data correlate highly with our benchmark measure throughout the 2010 to 2017 period for which data are currently available, although we find some evidence to suggest that they understate the concentration of poverty in the state's highest-poverty schools. These findings suggest that the validity of FRPL data depends on whether all students enrolled in schools that provide free meals to all students are counted as enrolled in FRPL. While we therefore conclude that the FRPL CCD data we analyze for Oregon public schools provide a reasonably valid measures of school poverty, we cannot address whether this might change in the years to come.

How, then, should policymakers and researchers who lack access to detailed household earnings data measure the concentration of poverty in public schools? Our findings indicate that school-level measures of the proportion of students who directly certify for school meals via enrollment in SNAP and other means-tested federal programs provide highly valid measures of school poverty. We find that temporal trends in SNAP enrollment rates parallel trends in our benchmark measure of student poverty. Further, we find that these two measures consistently

correlate quite highly across the time period for which we have data. Finally, we find that the correspondence between SNAP enrollment rates and our benchmark measure holds equally well for Oregon public schools across the poverty distribution. While these measures are not currently publicly available for all U.S. public schools, a growing number of states make them available via the CCD and all states have the capacity to do so. Notably, SNAP enrollment reflects a different family income threshold than FRPL enrollment, so switching from one measure to the other requires attention to the corresponding level change in poverty rates.

We find similarly encouraging evidence regarding the use of the Urban Institute's MEPS measure. We note, however, that the MEPS measure is designed as an adjustment for publicly-available data, rather than an alternative source of data. In the Oregon context, MEPS is calculated based on the relatively high-quality FRPL data reported in the CCD. It is not clear whether the MEPS measure would perform as well if it were calculated based on lower quality data in other states. Conversely, it is possible that the MEPS approach could help further improve the validity of school poverty measures in contexts where direct certification data might not be as strong of a measure.[20]

Our analyses proceed from the assumption that compensatory school policies intend to target resources at schools based on their enrolled concentration of students from low-income families. While this focus is an explicit component of many educational policies, including Title I of the ESEA and many state and local school finance plans, we recognize that it is a policy choice. In the future it may be desirable to broaden our view, redirecting attention from the dichotomous and unidimensional notion of student poverty and toward richer, more multidimensional conceptions of student advantage and disadvantage. More sophisticated

[20] For example, Oregon has a relatively high SNAP take up rate, and it is possible that the direct certification measure might not fare as well in states where this was not the case.

measures of student household economic status might consider household earnings as a continuous variable or incorporate local or regional cost of living data to better approximate household spending power. Further, in light of growing evidence regarding the role of wealth in the intergenerational reproduction of educational advantage (Hallsten and Pfeffer, 2017; Pfeffer and Killewald, 2018), policymakers and researchers may build wealth into their consideration of household economic status. Alternatively, researchers and policymakers may attempt to broaden their view to consider non-economic factors such as household composition, educational attainment, occupational status, and availability of learning opportunities. Our analyses are not designed to speak to these broader constructs, but future research should.

What our research makes clear, however, is that some commonly used measures of student poverty are considerably better than others. And while there is growing concern over the quality of poverty measurement in education – concerns that we show are well-founded – it is also the case that direct certification data provides a perhaps surprisingly strong measure that could be widely implemented.

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Table 1. List of measures of school poverty used in analyses

Name	*Source*	*Measure*	*Notes*
Benchmark measures			
Benchmark (SNAP/IRS 3 year)	Census (IRS/SNAP) 2009-10 through 2016-17	Proportion enrolled in SNAP at any point during relevant school year or with documented family income below <1.85 poverty	Students are designated low income if they (a) are enrolled in SNAP at any point during relevant school year or (b) have family income less than 1.85x poverty in their family's most recent IRS 1040 filling (within prior three tax years)
Benchmark (IRS 1 year)	Census (IRS) 2009-10 through 2016-17	Proportion with documented family income below <1.85 poverty	Students are designated low income if they have family income less than 1.85x poverty in their family's IRS 1040 from the most recent calendar year
Traditional FRPL enrollment measures			
CCD FRPL	NCES - CCD 2009-10 through 2016-17	Proportion enrolled in FRPL	Based on state's reports to the NCES
ODE Economic Disadvantage	Oregon Department of Education 2009-10 through 2016-17	Proportion flagged as "economically disadvantaged" based on NSLP enrollment and other factors	Other factors include children experiencing homelessness as defined by McKinney-Vento Homeless Assistance Act. In the 2016-2017 school year, all students flagged as "economically disadvantaged" if they attend schools that provide free lunch to all regardless of economic status
Direct certification measure			
SNAP	Census (SNAP) 2009-10 through 2016-17	Proportion enrolled in SNAP (proxy for direct certification)	Received any SNAP benefits during school year (e.g., for 2017, enrollment between August 2016 and July 2017), excluding students who only receive SNAP via summer school-based program
Alternative candidate measures			
ACS Block Group	ACS 2009-10 through 2016-17	Proportion <1.85 poverty in student's block group	Defined as the proportion of children ages 5-17 in households with income below 1.85x poverty in the student'sblock group
MEPS	Urban Institute 2013-14 through 2018-19	Predicted poverty rate from SAIPE and CCD	This measure uses a linear probability model to adjust school-level estimates of economic disadvantage based on average household poverty rates for families with children in the districts in which schools are located
NCES-SNP	NCES 2016-17	Income-to-poverty ratio	Based on the average income of households in the immediate vicinity of the school using ACS data
Transformed NCES-SNP	NCES and author calculations 2016-17	Proportion <1.3 poverty	Transformation of the SNP measure designed to capture the share of students at or below 130 percent of the federal poverty line for each school (see Fazlul et al., 2021)

Source: 2010-2017 ODE, IRS 1040, OR SNAP
DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.

Table 2. Enrollment weighted descriptive statistics of analytic sample

	Analytic sample	
	Mean	*Std. Dev.*
Benchmark measures		
Benchmark (SNAP/IRS 3 year)	0.506	0.185
Benchmark (IRS 1 year)	0.433	0.171
Traditional FRPL Enrollment Measures		
CCD FRPL	0.522	0.211
ODE Economic Disadvantage	0.539	0.242
Direct certification measure		
SNAP*	0.345	0.164
Alternative candidate measures		
ACS Block Group*	0.327	0.093
Urban Institute MEPS[+]	0.176	0.075
NCES-SNP (Income-to-poverty ratio)+	304.9	140.5
Transformed NCES-SNP[+]	0.399	0.106
Demographic characteristics		
Proportion Black	0.024	0.048
Proportion White	0.641	0.191
Proportion Hispanic	0.221	0.174
Proportion Multi-ethnic	0.048	0.031
Proportion American-Indian	0.016	0.041
Proportion Asian	0.049	0.063
Proportion ESL	0.227	0.219
School Enrollment	733.5	520.2
N (School-years)	8500	

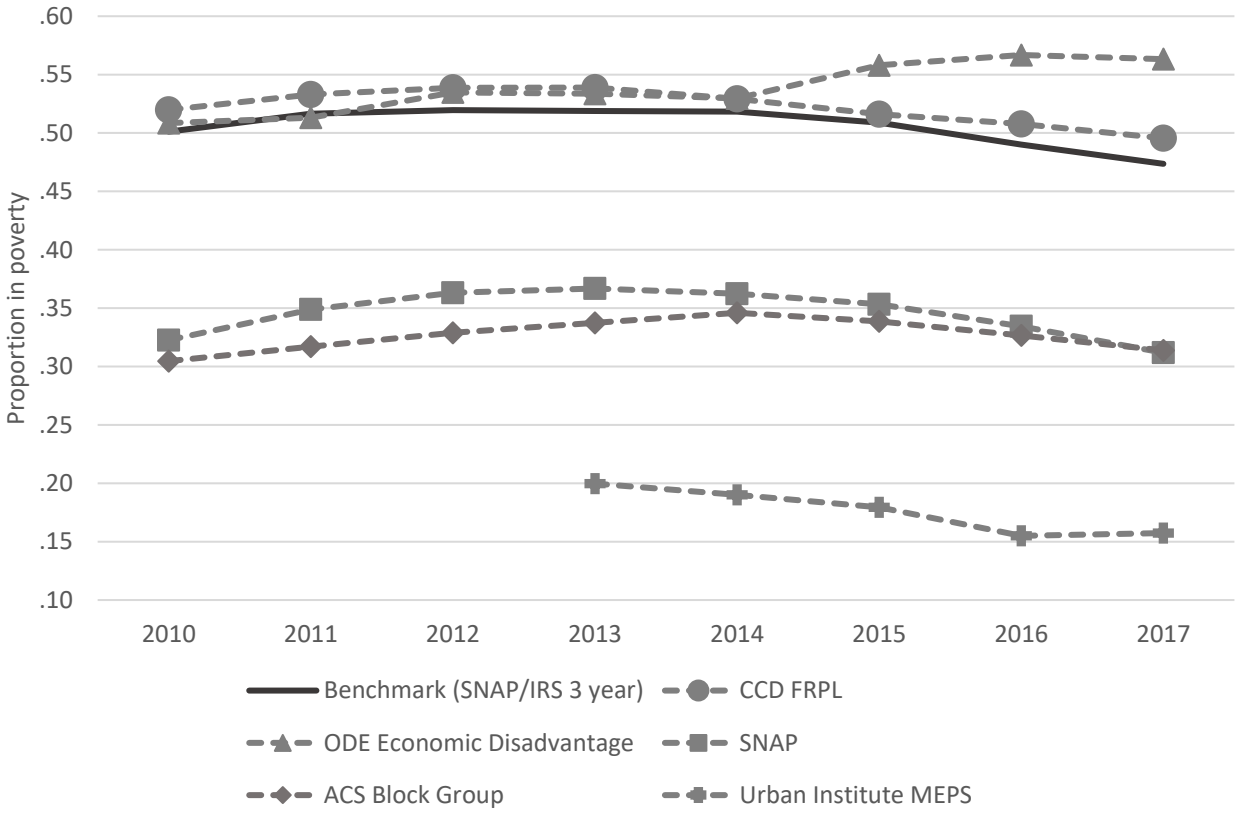
Source: 2010-2017 ODE, IRS 1040, OR SNAP
DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.

Notes. Statistics are enrollment weighted.

*Available for >99% of school-year observations in the full sample, but missing for a small number of school-years. Measures with suppressed values in the full sample have many more missing values. By construction, all measures are available for all schools in the longitudinal sample.

[+]MEPS measures are available 2013-2017; the SNP measure is available in 2017 only.

Figure 1. Trends in measures of student poverty in Oregon across measures, 2010-2017



Source: ODE, IRS 1040, OR SNAP
DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.

Table 3. Correlations between benchmark measure and candidate measures, 2016-2017 school year

	Benchmark (SNAP/IRS 3 year)	*Benchmark (IRS 1 year)*
Benchmark measures		
Benchmark (SNAP/IRS 3 year)	1	
Benchmark (IRS 1 year)	0.991	1
Traditional FRPL Enrollment Measures		
CCD FRPL	0.934	0.923
ODE Economic Disadvantage	0.863	0.859
Direct certification measure		
SNAP	0.957	0.942
Alternative candidate measures		
ACS Block Group	0.796	0.796
Urban Institute MEPS	0.935	0.929
NCES-SNP (Income-to-poverty ratio)	-0.694	-0.679
Transformed NCES-SNP	0.661	0.653
N Schools	1,100	

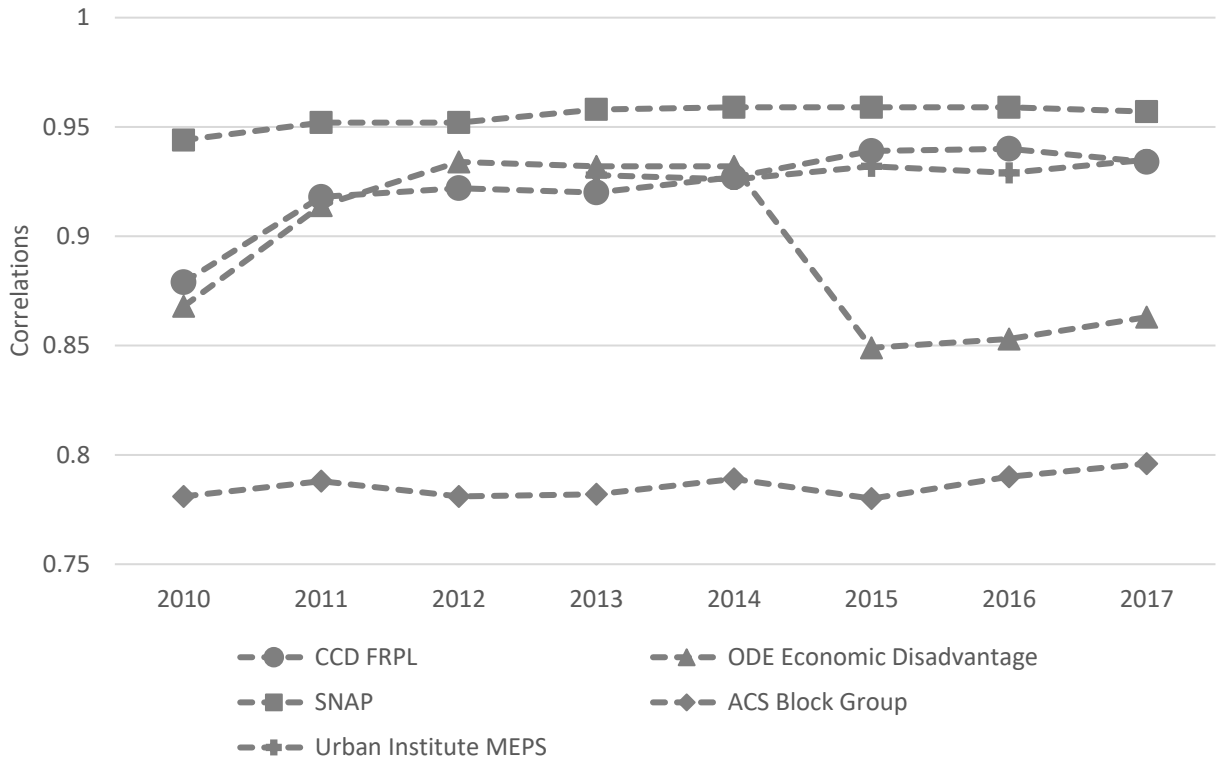
Source: ODE, IRS 1040, OR SNAP
DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.

Note. Statistics are not enrollment weighted.

Statistics include schools from the final year of the analytic sample (2016-2017) for which there is coverage from all measures. We lose approximately 300 schools with small enrollments driven in large part to whether schools are represented in the NCES data sources like the CCD and SNP measures. The analytic sample offers a reasonable snapshot of the universe of schools in the ODE data (see Appendix Table 1).

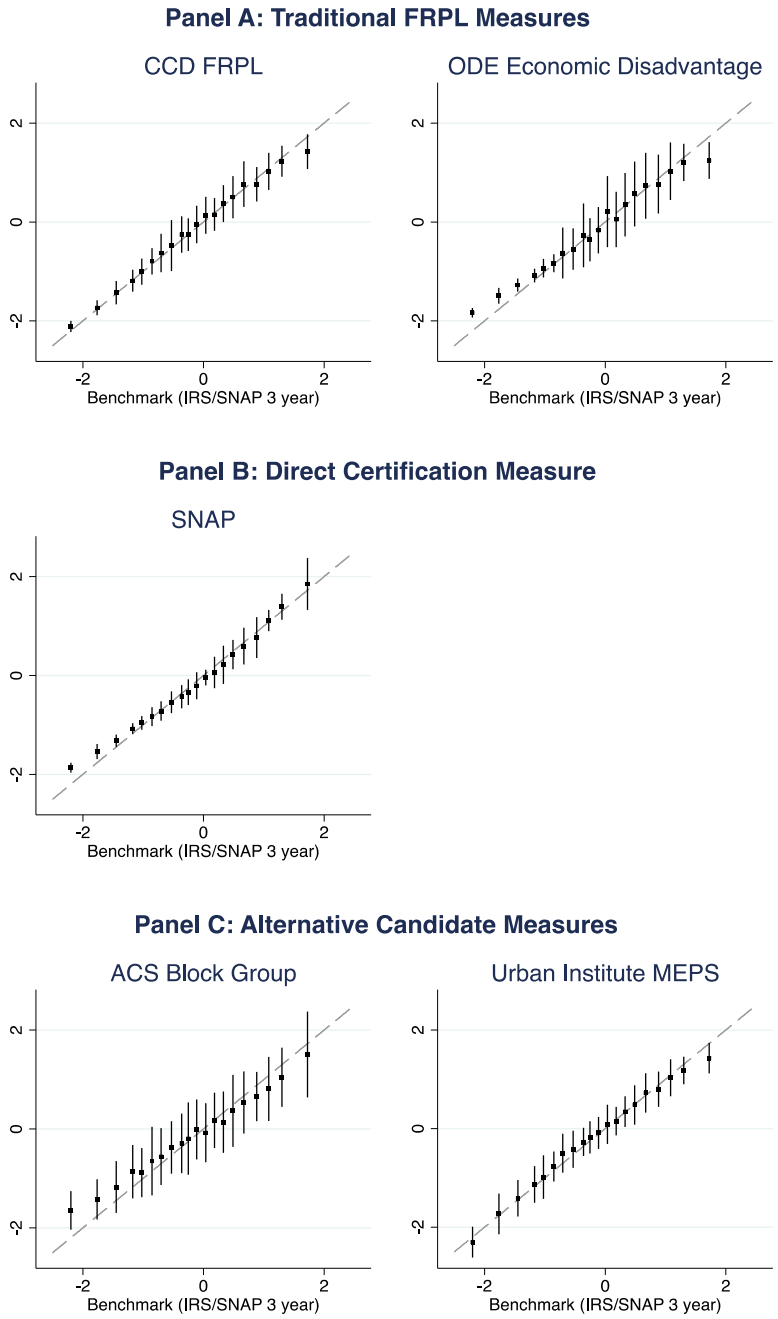
The Urban Institute also released a MEPS modified measure which they recommend to be used for analyses in large school districts with large school enrollments. Because we are conducting a state-wide analysis with schools of varying enrollment sizes, we do not focus on the MEPS modified measure in the paper. Consistent with Urban Institute's recommendation, the MEPS modified measure has a lower correlation with the benchmark than the MEPS, with a correlation of .865.

Figure 2. Trends in the correlation between the benchmark measure and candidate measures, 2010-2017



Source: ODE, IRS 1040, OR SNAP
DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.

Figure 3. Relationship between candidate measures and benchmark across the distributions



Source: ODE, IRS 1040, OR SNAP
DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.
Note. Data is from academic year 2016-2017.

Appendix Table 1. Enrollment weighted descriptive statistics of analytic sample and the universe of schools in ODE data, 2009-10 through 2016-17

	Universe of schools in ODE data		Analytic sample	
	Mean	*Std. Dev.*	*Mean*	*Std. Dev.*
Benchmark measures				
Benchmark (SNAP/IRS 3 year)	0.508	0.186	0.506	0.185
Benchmark (IRS 1 year)	0.435	0.172	0.433	0.171
Traditional FRPL enrollment measures				
CCD NSLP	-	-	0.522	0.211
ODE Economic Disadvantage	0.530	0.245	0.539	0.242
Direct certification measure				
SNAP*	0.346	0.165	0.345	0.164
Alternative candidate measures				
ACS Block Group*	0.327	0.092	0.327	0.093
Urban Institute MEPS[+]	-	-	0.176	0.075
NCES-SNP (Income-to-poverty ratio)+	-	-	304.9	140.5
Transformed NCES-SNP[+]	-	-	0.399	0.106
Demographic characteristics				
Proportion Black	0.025	0.052	0.024	0.048
Proportion White	0.646	0.190	0.641	0.191
Proportion Hispanic	0.216	0.173	0.221	0.174
Proportion Multi-ethnic	0.049	0.032	0.048	0.031
Proportion American-Indian	0.016	0.041	0.016	0.041
Proportion Asian	0.048	0.062	0.049	0.063
Proportion ESL	0.220	0.219	0.227	0.219
School Enrollment	720.5	564	733.5	520.2
N (School-years)	12000		8500	

Source: ODE, IRS 1040, OR SNAP
DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.

Notes. Statistics are enrollment weighted.

*Available for >99% of school-year observations in the full sample, but missing for a small number of school-years. Measures with suppressed values in the full sample have many more missing values. By construction, all measures are available for all schools in the analytic sample.

[+]MEPS measures are available 2013-2017; the SNP measure is available in 2017 only.